QUALITY ASSURANCE · TAILORED FOR REGANT

REGANT QA Guide

The Generic System QA Guide applied to REGANT, the Decision Support System of the Tubungan Teachers' Multi-Purpose Cooperative (TTMPC): its eight portals, the loan lifecycle from application to full payment, membership, savings, CBU, ISC, receipts, and the three analytics engines (credit risk, loan demand forecast, MIGS).

System
REGANT DSS for TTMPC
React + Vite · FastAPI · Supabase

Covers
Access, loan lifecycle, other flows, monitoring, charts, I/O, cross-cutting

Based on
Generic System QA Guide
(same IDs and result codes)

BEFORE YOU TEST

How to use this guide

This is the generic guide with every placeholder replaced by REGANT's own terms, screens, statuses and limits, taken from the code and the project documents (CLAUDE.md, WHAT_IFS_AND_CONSTRAINTS.txt, RECEIPT_SYSTEM_PLAN.md, ISC_DIVIDEND_PLAN.md, ACCOUNT_MANAGEMENT_TERMINATION_PLAN.md). Case IDs keep the generic prefixes. Cases that cannot apply to REGANT are kept and marked N/A with the reason, so totals stay comparable.

1. Placeholders, as they apply to REGANT

PLACEHOLDER	IN REGANT
<record>	Mainly a **loan application** (`loans` row, keyed by control number). Secondary records: membership application, loan payment, savings transaction, CBU deposit, ISC posting, renewal-override request, receipt.
<unit>	TTMPC is one cooperative with no branches, so data scope is **the individual member** (a member sees only their own loans, savings, CBU, receipts) and **the role** (each staff portal sees only its own screens). "Unit A / Unit B" in the generic guide becomes **Member A / Member B**.
<approver role>	Loans: Bookkeeper (evaluates and recommends) → BOD (only Consolidated/KOICA over ₱500,000) → Manager (approves) → Treasurer (clears for release) → Cashier (releases). Membership: BOD. Payments and withdrawals: Bookkeeper validates. Renewal override: Bookkeeper (not their own request). ISC reversal: Manager (not their own posting).
<balance>	Loan outstanding balance and amortization schedule; savings balance; CBU / share capital (`member.share_capital_amount`); Treasurer vault cash on hand; grocery credit balance.

2. System profile

FIELD	VALUE
System name	REGANT: Decision Support System for TTMPC
Version / build	Git commit: _____ (record `git rev-parse --short HEAD`)
Test environment URL	Frontend `http://localhost:5173` (or the staging deploy); API `http://127.0.0.1:8000`; API docs `/docs`
Database or data copy used	Supabase project: _____ (use a Supabase branch or a copy, not the live cooperative data)
Test period	_____
Roles	Member, Non-member (KOICA/ABFF), Cashier, Bookkeeper, Manager, Treasurer, BOD (incl. the account-admin BOD), Secretary
Scope units	Individual member (own data); role (own portal)
Core process (one line)	Member applies for a loan → Bookkeeper evaluates (MIGS, credit risk) → BOD if Consolidated > ₱500,000 → Manager approves → Treasurer clears → Cashier releases → Cashier records payments, Bookkeeper validates → Fully paid or renewed
External services	Supabase Auth / Postgres / Realtime; Resend (email); POS webhook (grocery); local ML models (credit risk `ttmpc_risk_model.pkl`, SARIMAX demand models)
Supported browsers and devices	Chrome and Edge (current) on desktop for staff; a phone browser for the Member portal (it has a bottom navigation)

FIELD	VALUE
Reference documents	CLAUDE.md, WHAT_IFS_AND_CONSTRAINTS.txt, DEMO_GUIDE.md, RECEIPT_SYSTEM_PLAN.md, ISC_DIVIDEND_PLAN.md, ACCOUNT_MANAGEMENT_TERMINATION_PLAN.md, the TTMPC loan policies (Consolidated, Emergency, Bonus, ABFF/KOICA)

3. Scope and structure

PART	WHAT IT COVERS	PREFIX
Before you test	Environment, accounts, result codes, severity, entry and exit criteria, access and role checks.	AC
A. Loan lifecycle	Prerequisites → apply → Bookkeeper evaluation → BOD (high value) → Manager → Treasurer → Cashier release → repayment → renewal, restructure and closure; then reconciliation.	PF0–PF8, RC
A2. Other flows	Membership, savings, CBU, ISC, account termination and payout, grocery / POS.	MB, SV, CB, IS, TM, GR
B. Monitoring	The eight portals' dashboards, queues, lists, search, drill-down, realtime updates.	MO
C. Visualizations	Universal chart checks and the REGANT chart catalog (Recharts dashboards, demand forecast, credit risk).	VZ
D. Inputs and outputs	Field validation, audit log, notifications, emails, receipts, PDFs.	IN, OU, RP
E. Cross-cutting	Dialogs, responsive layout, accessibility, browsers, performance, network, security, data integrity, time and locale.	XC, RL, AX, BC, PE, NW, SE, DI, TL
Appendices	Status map, business rules, boundary values, documentation differences, defect report, sign-off, glossary.	—

4. Recording results

CODE	MEANING
P Pass	The expected result happened exactly as written, including messages, statuses, and every updated figure (₱ balance, schedule row, receipt, audit entry) named in the case.
F Fail	Any part did not happen, or something extra happened: a wrong peso amount, an extra loan or receipt, or another member's data visible.
B Blocked	Could not run because an earlier step failed or a prerequisite (account, seed loan, Resend key, model file) was missing. Note the blocking case ID.
N/A	Does not apply to REGANT, this build or environment. Write the reason.

SEVERITY

SEVERITY	USE WHEN	REGANT EXAMPLES
Critical	Money or balances are wrong or lost, totals do not reconcile, a security hole, or a user sees or changes another member's data or another role's actions.	A payment is validated twice; a loan is released twice; Member B opens Member A's receipt; a terminated member can still sign in.
High	A core workflow cannot be completed and there is no workaround.	Manager cannot approve a valid loan; Cashier cannot release; SOA PDF never downloads.
Medium	A wrong display, filter, or message, with a workaround.	A dashboard chart ignores the year filter; an error names the wrong field.
Low	Cosmetic or wording issues.	A misaligned button; inconsistent "CBU" / "Share Capital" wording.

5. Test environment and data

Never test against the live cooperative database
REGANT writes straight to Supabase, and Resend sends real emails. Use a Supabase branch or a copied project, and set member emails to inboxes you control. Test loans in the live database distort the demand forecast, MIGS scores and the Bookkeeper Reports, and receipts can never be deleted (by design).

- **Two servers.** Start the API (`uvicorn main:app --reload --host 127.0.0.1 --port 8000` in `src/server`) and the frontend (`npm run dev`). Hit `GET /api/analytics/demand/forecast` once to warm the SARIMAX models.
- **One browser profile per role.** Sessions use `sessionStorage`, so each tab is its own session. Use separate profiles or private windows to keep each role signed in.
- **Known starting data.** Before starting, note each test member's savings balance, CBU, loan outstanding balance and the vault balance.
- **Date-dependent rules.** The Bonus loan window (May and November only) is enforced by a database trigger on `now()`; the 6-month renewal rule, 90-day delinquency and the 3-month penalty grace all use dates. The business time zone is Asia/Manila (UTC+8). Check the DB time zone.
- **SQL migrations applied.** Confirm the receipts (01–06), account-management (part A–C), renewal-override, ISC v2 (01–06) and disbursement-confirmation SQL files are applied, or the related cases are Blocked.
- **Clean-up.** Keep a list of the control numbers, membership IDs and receipt numbers you create.

6. Accounts to prepare

ACCOUNT	PURPOSE	HOW TO PREPARE
Account-admin BOD	Uses Account Management: role changes, termination.	`member_account.can_manage_accounts = true`
BOD (regular)	Member approval, high-value loan decision, loan policies; must be refused Account Management.	Role `bod`, no admin flag
Secretary	Attendance, General Assembly, Records (view-only).	Role `secretary`
Manager	Loan approval, credit risk view, ISC reversal.	Role `manager`
Bookkeeper × 2	Evaluation, MIGS, payment validation. Two are needed for the "not your own request" checks (renewal override, ISC post vs reverse).	Role `bookkeeper`; one should also be a member
Treasurer	Release clearance, vault, accounting.	Role `treasurer`
Cashier	Payments, deposits, withdrawals, release, CBU, membership fees.	Role `cashier`
Member A (MIGS)	Main flow. Active loan with ≥ 6 recorded payments, savings and CBU.	MIGS score ≥ 50
Member B (Non-MIGS)	Scope checks; active loan with < 6 payments (renewal block).	MIGS score < 50
Exiting / terminated member	Status gates: no new loans, no CBU deposits.	Terminate through Account Management
Pending applicant	Submitted membership form, not yet approved.	Sign Up + Membership Form, leave pending
Non-member (KOICA/ABFF)	Non-member loan path.	Via `/Non_Member` → `/Koica_Forms`
Staff with temporary credentials	`AccountSetupGate` forces email/password setup.	Provisioned with a temporary email/password

SEED THE MINIMUM DATA

- Loan types with rates in `loan_types` and a row per type in `loan_fee_policies` (BOD Loan Policies).
- At least one loan in each loan status of Appendix A1, including one Consolidated loan above ₱500,000.
- One payment `pending_bookkeeper`, one savings withdrawal `pending_verification`.
- A member with the longest names/addresses allowed, to test layouts and PDFs.
- At least 12 months of loan history so the dashboards and demand forecast have data.

7. Roles, screens, and scope (role matrix)

ROLE	SCOPE	SCREENS (AS BUILT)	CAN CREATE / APPROVE / EXPORT
Member	Own data only	/member-dashboard, /member-loans, /member-apply-loans, /member-savings, /member-statement-of-account (Loan · Savings · Transactions · Capital Build-Up), /members-profile; loan forms /Consolidated_Loan, /Emergency_Loan, /Bonus_Loan	Apply for loans; request renewal override; download SOA / savings / CBU PDFs and own receipts
Non-member (KOICA)	Own application	/Non_Member, /Koica_Forms, /loan_kiosk	Apply for a KOICA/ABFF loan
Cashier	All members (transactions)	/Cashier_Dashboard, /Cashier_Payments, /Cashier_Savings, /Cashier_Withdrawals, /Cashier_Disbursement, /Cashier_CBU, /Cashier_CBU_Deposit/:memberId, /Cashier_CBU_Exit_Payout, /Cashier_MembershipPayments, /Cashier_Grocery, /cashier-audit-log	Record payments, deposits, withdrawals, CBU, membership fees, exit payouts; release loans. Receipt pop-up only, no reprint
Bookkeeper	All members	/dashboard, /loan-application, /bookkeeper-loan-approval(/:id), /manage-loans, /manage-member, /migs, /migs-evaluate, /bookkeeper-credit-risk, /payments, /bookkeeper-savings-transactions, /bookkeeper-savings-accounts, /bookkeeper-cbu, /bookkeeper-isc, /bookkeeper-isc-journal, /bookkeeper-renewal-overrides, /grocery, /grocery-ledger, /reports, /records, /accounting, /audit-trail, /bookkeeper-receipts, /bookkeeper-loan-ledger/:loanId	Recommend loans; validate payments and withdrawals; run MIGS; post ISC; approve overrides; Executive PDF, SOA
Manager	All members	/manager-dashboard, /loan-approval(/:id), /manager-credit-risk, /manager-manage-loans, /manager-manage-member, /manager-savings-accounts, /manager-reports, /manager-audit-log, /manager-receipts, /manager-loan-ledger/:loanId	Approve / reject / return loans; reverse ISC postings (not own)
Treasurer	All members	/Treasurer_Dashboard, /treasurer-approval(/:id), /disbursement, /treasurer-payments, /treasurer-vault, /treasurer-accounting, /treasurer-savings-accounts, /treasurer-audit-log, /treasurer-receipts	Clear for release, reschedule, reject; vault entries
BOD	All members	/BOD-dashboard, /member-approvals(/:id), /bod-loan-approvals, /bod-loan-approval/:id, /bod-manage-member, /bod-manage-loans, /bod-loan-policies, /bod-audit-log, /bod-receipts, /bod-account-management (admin BOD only)	Approve members; decide loans > ₱500,000; edit loan/fee policies; account management (admin)
Secretary	Membership records	/Secretary_Attendance, /Secretary_General_Assembly, /Secretary_Records, /secretary-record-details/:id	Attendance; records are view-only; no receipts

8. Entry and exit criteria

START TESTING WHEN	FINISH TESTING WHEN
• **Both servers run; the commit hash is recorded.** • **Smoke test passes: each role signs in and opens its home screen.** • **Accounts and seed data are ready; the System profile is filled in.** • **All required SQL migrations are applied in the test database.** • **WHAT_IFS_AND_CONSTRAINTS.txt and the loan policies are at hand.**	• Every case has P, F, B, or N/A with a reason. • No open Critical or High defects; Medium defects have an agreed fix plan. • Fixed defects are retested and affected cases rerun. • The test summary is signed in Appendix A6.

9. Access and role checks

Run these before the loan lifecycle. Protected pages use `RequireRole` in the browser; the FastAPI endpoints and Supabase Row-Level Security must refuse the same actions on the server.

ID	TEST STEPS	EXPECTED RESULT	RESULT
AC-01	Sign in with each approved account (staff via `/Login`, members via `/memberlogin`).	Each lands on its own home screen (e.g. Cashier → /Cashier_Dashboard); the header shows the right name and role.	
AC-02	Sign in with a wrong password, then with an email that has no account.	Same generic message for both; it does not reveal whether the email exists.	
AC-03	Fail to sign in repeatedly (11+ times).	Supabase Auth rate limiting slows or refuses further attempts; record the exact behaviour (REGANT documents no lockout count; note it in A4).	
AC-04	Sign Up and submit the Membership Form as a new applicant, then try the Member portal before BOD approval.	No member data screen opens; a pending-approval message is shown.	
AC-05	As BOD, approve the applicant at /member-approvals/:id after training and fee payment; sign in again with it.	The applicant becomes a member (`application_status = 'Member'`, membership ID assigned) and the Member portal opens. The approval is in the BOD audit log with actor and time.	
AC-06	As BOD, reject a second pending applicant.	Status Rejected; the applicant cannot open member screens.	
AC-07	On Sign Up / Role Selection, check which roles can be chosen; then send a staff role (e.g. `role: 'manager'`) by editing the request.	Only applicant/member paths are offered; the server or RLS refuses a self-assigned staff role.	
AC-08	Signed in as a Member, type /dashboard, /loan-approval, /Treasurer_Dashboard, /bod-account-management.	Redirected by RequireRole; no staff data shows, not even briefly.	
AC-09	Signed in as Member B, look for Member A's loans, savings, CBU, receipts in every list, the SOA and the downloaded PDFs.	None appear; every list and file is limited to Member B.	
AC-10	As Member B, open one of Member A's receipts by putting Member A's receipt number in `?receipt=`, and call the receipt API with it.	404 (the receipt's existence is not revealed); no data shown.	

ID	TEST STEPS	EXPECTED RESULT	RESULT
AC-11	Forgot password at /forgot-password: request the code, verify at /verify-reset-code, set a new password at /reset-password on another device. Reuse the code.	The reset works; the reused or expired code is refused.	
AC-12	Set a new password shorter than 8 characters, and with a mismatched confirmation.	Refused with the rule shown; works only when all rules pass.	
AC-13	Leave a session idle for a long period (e.g. 1 hour+), then act.	Asked to sign in again or the token refreshes silently, as documented; record which. No action completes without a valid session.	
AC-14	Sign out, then press Back.	No data is shown until someone signs in again.	
AC-15	As the account-admin BOD, terminate a member who is signed in elsewhere.	Their next API call fails and sign-in is refused (the Supabase auth user is banned); the termination is logged with a reason.	
AC-16	Reactivate (or close, per the termination plan) the account from AC-15.	It returns to the documented status; a closed member's login is deleted and personal contact fields are cleared, but name, membership ID, date of birth and financial history are kept.	
AC-17	As account-admin BOD, change a staff member's role (e.g. Cashier → Bookkeeper); use the account.	New portal on next sign-in; the old portal's routes are refused. The Member portal stays open to them as a member.	
AC-18	Sign in with a staff account that has temporary credentials.	AccountSetupGate forces setting a real email and password before any portal screen.	
AC-19	As Cashier, replay a Manager-only action (e.g. update `loans.loan_status` to 'to be disbursed' via the Supabase client, or call `POST /api/bod/loan-decision`).	Refused by RLS / the endpoint (403); the loan is unchanged. *Known risk:* loan status transitions are written from the browser (WHAT_IFS: “no FastAPI transition guard”), so test this carefully.	
AC-20	Sign in to the same account in two tabs and on two devices.	Each tab has its own session (sessionStorage); signing out in one tab behaves as documented in the other.	
AC-21	Without signing in, open /dev/pos-simulator and call `GET /api/dev/pos-simulator/members` and `POST /api/dev/simulate-pos-event`.	Not reachable on a deployed build (or refused without auth). *Known risk:* these dev routes have no auth guard.	
AC-22	As a non-account-admin BOD, open /bod-account-management and call `/api/admin/accounts/*`.	The page says access is restricted; the API returns 403.	

PART A · LOAN LIFECYCLE

The core process, stage by stage

In REGANT the main record is a loan application. It moves through five staff roles and, once released, moves money: the member's loan balance, a CBU deduction, the vault, and a receipt. Run each stage on the real screens.

Stage index

STAGE	WHAT HAPPENS	ACTOR	STATUS AFTER	PREFIX
0	Prerequisites: loan types, fee policies, MIGS scores, member profile	BOD / Bookkeeper	—	PF0
1	Member applies (Consolidated, Emergency, Bonus; KOICA for non-members)	Member	pending	PF1
2	Bookkeeper evaluation: MIGS, credit risk, co-makers, collateral	Bookkeeper	recommended for approval / recommended for bod approval	PF2
3	BOD decision (Consolidated/KOICA > ₱500,000 only)	BOD	recommended for approval / bod rejected	PF3
4	Manager review: approve, reject, or return for revision	Manager	to be disbursed / rejected / revision_requested	PF4
5	Treasurer clearance: release, reschedule, or reject	Treasurer	ready for disbursement / pending rescheduling / rejected	PF5
6	Cashier releases the loan (money moves)	Cashier	released	PF6
7	Repayment: Cashier records, Bookkeeper validates	Cashier → Bookkeeper	payment validated; loan fully paid at ₱0	PF7
8	Renewal, restructure / reschedule, closure	Member, Bookkeeper, Manager	renewed (old loan paid off) / rescheduled / fully paid	PF8
—	End-to-end reconciliation	QA assessor	—	RC

Balances move only at stages 6, 7 and 8
Stages 1–5 change only `loan_status`. Money moves at release (stage 6: loan balance, CBU deduction, vault, receipt), at payment validation (stage 7: schedule, balance, receipt) and at renewal (stage 8: the new loan pays off the old one). A receipt at any earlier stage is a defect.

0 Prerequisites: loan settings and member data

BOD · Bookkeeper

Where: BOD › Loan Policies (/bod-loan-policies); Bookkeeper › MIGS (/migs); member profile

Rates, fee policies and MIGS classifications must exist before a loan can be computed. Only BOD may edit loan/fee policies (RLS `loan_fee_policies_write_bod`).

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF0-01	Open BOD Loan Policies and check each loan type (Consolidated, Emergency, Bonus, KOICA) has a rate and fee policy.	Every type has a configured rate, service fee, CBU rate and insurance; none shows blank.	
PF0-02	As BOD, change a fee value (e.g. insurance per thousand); then enter a whole number where a fraction is expected (2 instead of 0.02).	The valid change saves; the out-of-range value is refused or warned (WHAT_IFS notes no range CHECK exists: record what happens).	

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF0-03	Delete or empty a loan type's fee-policy row, then compute that loan type.	An explicit "policy missing" error, not a silent ₱0 CBU retention.	
PF0-04	As Bookkeeper / Manager, try to edit loan policies (screen and a direct Supabase update).	No control; RLS refuses the write.	
PF0-05	As Bookkeeper, run `POST /api/migs/recompute-all` from /migs; open a member's MIGS breakdown.	7 criteria totalling 100; score ≥ 50 = MIGS; the change is reflected in the loan multiplier for new applications.	

1 Member applies for a loan

Member

Where: Member › Apply for Loan (/member-apply-loans) › Consolidated / Emergency / Bonus form; Non-member › /Koica_Forms

The form computes the loan, checks eligibility through `get_loan_eligibility`, and submits a `loans` row with status `pending`.

INPUTS

FIELD	TYPE / OPTIONS	RULES
Loan type	Consolidated, Emergency, Bonus (members); KOICA/ABFF (non-members)	Consolidated/Bonus: bona fide members only. Bonus: DepEd only, May and November only (DB trigger).
Loan amount	Money (₱)	> 0. Emergency ≤ ₱20,000. Ceiling = CBU × 5 (MIGS) or × 3 (Non-MIGS).
Term	Select	Consolidated 12/24/36/48/60 months; Emergency 6/12 months.
Application type	New / Renewal	Renewal needs 6 validated payments on the active loan, or an approved override.
Co-makers, collateral	Member search; collateral type co_maker / cbu_shares / chattel / real_estate / atm_hold / other	Values ≥ 0; a member can attach collateral only to their own loan (RLS).

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF1-01	Apply for a Consolidated loan with valid values.	Saved with a confirmation; appears in Member Loans and in the Bookkeeper's /loan-application queue as `pending`, with the computed amortization, fees and net proceeds.	
PF1-02	Try to submit with each required field empty in turn.	Blocked; the message names the field; other entries are kept.	
PF1-03	Apply for a second loan of the same type while one is active (not a renewal).	Blocked with the eligibility reason (active loan ID); no second row.	
PF1-04	Double-click Submit.	Exactly one loan row; the button is disabled while saving.	
PF1-05	Fill in the form, then leave the page.	Asked to confirm discarding; nothing saved.	
PF1-06	Apply as a member with missing net pay / classification / employment type.	400 with the missing fields listed.	
PF1-07	Emergency loan of ₱20,000.01; Consolidated with term 18.	422 with the rule ("cannot exceed 20,000"; allowed terms).	
PF1-08	Apply for a Bonus loan outside May/November (and as a non-DepEd member).	Refused, even by a direct Supabase insert (trigger `enforce_bonus_loan_window`).	

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF1-09	Apply as an exiting / terminated member, and as a Non-MIGS member for a Consolidated loan.	403 with the member's status / "bona fide members only".	
PF1-10	Open the audit log / loan notifications.	One create entry and one bell notification for the Bookkeeper.	

2 Bookkeeper evaluation

Bookkeeper

Where: Bookkeeper › Loan Application (/loan-application) › Loan Approval details (/bookkeeper-loan-approval/:id); MIGS (/migs-evaluate); Credit Risk (/bookkeeper-credit-risk)

The Bookkeeper reviews the application, MIGS score and credit-risk score, records co-makers and appraises collateral, then recommends.

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF2-01	Open a pending loan; recommend a Consolidated ≤ ₱500,000.	Status becomes `recommended for approval`; it leaves the Bookkeeper queue and appears in the Manager's /loan-approval.	
PF2-02	Recommend a Consolidated of ₱500,000.00 and one of ₱500,000.01.	₱500,000.00 → Manager; ₱500,000.01 → `recommended for bod approval` (BOD queue only; hidden from Manager).	
PF2-03	Recommend a loan that has declared collateral with no appraised value.	Blocked: "Please appraise all declared collateral items before sending to Manager."	
PF2-04	Open the Credit Risk score for the loan.	Probability and band come from `GET /api/risk/model-info` thresholds; missing inputs are shown as missing, not imputed.	
PF2-05	Change the recommended amount or co-makers, save, reload.	Changes kept; derived amortization updates.	
PF2-06	Two Bookkeepers recommend the same loan at the same time.	Only one status change applies; the second is told it was already handled.	
PF2-07	Open the loan's history / audit trail.	The evaluation shows who recommended it, when, and the remarks.	

3 BOD high-value decision

BOD

Where: BOD › Loan Approvals (/bod-loan-approvals › /bod-loan-approval/:id)

Only Consolidated (or KOICA) loans above ₱500,000 come here. The decision is made through `POST /api/bod/loan-decision`.

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF3-01	Approve with a resolution number and date.	Status `recommended for approval`; the loan enters the Manager queue.	
PF3-02	Approve without resolution no./date; reject without remarks.	422 before any write.	
PF3-03	Reject with remarks; then try to decide the same loan again.	`bod rejected` (terminal); the second decision is refused with 409.	
PF3-04	Call the endpoint as Manager / Treasurer / Cashier.	403.	

4 Manager review

Manager

Where: Manager › Loan Approval (/loan-approval › /loan-approval/:id)

The Manager sees the Bookkeeper's evaluation, MIGS and risk score, and approves (to be disbursed), rejects, or returns for revision.

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF4-01	Approve a recommended loan.	Status `to be disbursed`; approver and time recorded; member notified (bell + email); the loan appears for the Treasurer.	
PF4-02	Reject or return without remarks.	Blocked; remarks required.	
PF4-03	Reject with remarks.	`rejected`; the member sees the reason; the loan cannot continue.	
PF4-04	Return for revision.	`revision_requested`; it goes back to the Bookkeeper with the remarks; resubmitting returns it to the Manager.	
PF4-05	As a Manager who is also the applicant member, approve your own loan.	Not possible (record behaviour; segregation of duties).	
PF4-06	Open a loan that is still `recommended for bod approval` by its URL.	Not actionable by the Manager.	
PF4-07	Two Managers approve/reject the same loan at the same time.	Only the first decision applies.	
PF4-08	Approve after the member's status changed to exiting, or their CBU fell below the ceiling basis.	Warned or blocked as documented.	

5 Treasurer clearance

Treasurer

Where: Treasurer › Approval (/treasurer-approval › /treasurer-approval/:id)

The Treasurer clears an approved loan for release (`ready for disbursement`), parks it for rescheduling, or rejects it.

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF5-01	Clear a `to be disbursed` loan for release.	`ready for disbursement`; a disbursement row "Ready for Release" is created; it appears in Cashier Disbursement.	
PF5-02	Reschedule with remarks (the Awaiting tab).	`pending rescheduling`; it appears in the rescheduled-loans list.	
PF5-03	Reject or reschedule without remarks.	"Please provide remarks before confirming this action."	
PF5-04	Clear the same loan twice (refresh, Back, double-click).	Only one disbursement row.	
PF5-05	Clear a loan when the vault balance is lower than the net proceeds.	Warned or blocked (WHAT_IFS: undecided; record behaviour).	

6 Cashier releases the loan

Cashier

Where: Cashier › Disbursement (/Cashier_Disbursement)

Release calls `POST /api/cashier/disbursements/{loan_id}/disburse`: it writes a disbursement confirmation, moves the loan to `released`, creates the amortization schedule, deducts CBU retention (trigger) and issues a receipt.

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF6-01	Release a `ready for disbursement` loan with a reference number.	`released`; schedule rows created; CBU increases by the retention; receipt pop-up shows TTMPC-TR-YYYY-NNNNNN; the member sees the loan as active.	
PF6-02	Release with a reference number already used.	409 with the existing confirmation; nothing moves.	
PF6-03	Release a loan that is not `ready for disbursement` (direct request).	400; nothing written.	
PF6-04	Release with a stale session.	"Please sign in again", not a raw 422.	
PF6-05	Check the member's SOA, CBU tab, the vault and Bookkeeper Reports after release.	Before + change = after everywhere; net proceeds = principal − fees − CBU − insurance − old balance (renewal).	

7 Repayment

Cashier → Bookkeeper

Where: Cashier › Payments (/Cashier_Payments); Bookkeeper › Payments (/payments); Loan Ledger (/bookkeeper-loan-ledger/:loanId)

The Cashier records a payment (`pending_bookkeeper`); the Bookkeeper validates or rejects it. Only validation updates the schedule and issues a receipt.

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF7-01	Record a payment equal to one monthly due; validate it as Bookkeeper.	Schedule row Paid; outstanding balance falls by the principal portion; receipt issued at validation (not at recording).	
PF7-02	Record a partial payment.	Schedule row Partially Paid; the remainder stays due.	
PF7-03	Record ₱0, a negative amount, or negative penalty.	400.	
PF7-04	Validate the same payment twice.	409; no second receipt or balance change.	
PF7-05	Record a payment against a schedule that belongs to a different loan, or a loan with no schedules.	400.	
PF7-06	Pay the final balance.	Loan `fully paid`; further payments 409.	
PF7-07	Let a schedule pass 3 months unpaid (non-legacy loan); then a legacy loan.	Penalty 2%/month (1% Bonus) on remaining principal after the 3-month grace; legacy loans get no penalty.	
PF7-08	Reject a pending payment as Bookkeeper.	Rejected with reason; no balance change; no receipt.	

8 Renewal, restructure / reschedule, closure

Member · Bookkeeper · Manager

Where: Member Apply Loans + Renewal Override modal; Bookkeeper › Renewal Overrides (/bookkeeper-renewal-overrides); Treasurer reschedule list

A renewal is a **new** loan row that pays off the old one. A restructure/reschedule changes the **same** loan's term/schedule. Records are never deleted.

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF8-01	Member A (≥ 6 validated payments) renews the same loan type; release it.	New loan row; the old loan's remaining schedules become Paid; the old balance is deducted from net proceeds; active-loan count stays 1 for that type.	
PF8-02	Member B (< 6 payments) tries to renew.	Blocked; offered a renewal override request.	
PF8-03	Member B requests an override; Bookkeeper approves; Member B renews; tries a second renewal.	Approval unlocks one renewal for 30 days; it is consumed after the renewal; the second is blocked.	
PF8-04	A Bookkeeper who is also a member reviews their own override request.	Refused.	
PF8-05	Approve/reject an override that is not pending.	409.	
PF8-06	Submit a second restructure request for a loan that already has one pending; enter term or amortization ≤ 0.	Second request refused; ≤ 0 refused with 400.	
PF8-07	Approve a restructure.	The loan's own term/amortization and schedule change, not just the request status.	
PF8-08	Look for any delete action on loans, payments or receipts in any portal.	None; reversals are new entries; receipts cannot be edited or deleted (DB trigger).	

End-to-end reconciliation

ID	TEST STEPS	EXPECTED RESULT	RESULT
RC-01	Take one loan from application to fully paid, noting loan balance, CBU, savings, vault after each stage.	Balances change only at stages 6–8; final totals match a hand calculation from the policy (e.g. Consolidated 0.83%/month add-on).	
RC-02	Compare one figure (active loans, total CBU, collections this month) on the dashboards, Bookkeeper Reports, Manager Reports and the Executive PDF.	Identical for the same filters. Total CBU = sum of `member.share_capital_amount`.	
RC-03	Add up a member's ledger (SOA) for the period.	Opening + in − out = closing, for loan, savings and CBU.	
RC-04	Read the loan's status history.	Every change is allowed by Appendix A1; no skipped steps (e.g. pending → released).	
RC-05	Check what each role sees for the loan.	Matches the role matrix.	
RC-06	Read the audit log for the whole flow (each role's audit-log page).	One entry per action, in order, with the right actor.	
RC-07	Check notifications and emails for the whole flow.	Each actor gets exactly their notifications, once.	
RC-08	Run the flow for Member A and Member B at the same time.	Their balances stay independent.	

ID	TEST STEPS	EXPECTED RESULT	RESULT
RC-09	Count receipts for the flow.	One per release and one per validated payment; none for approvals, rejections or pending items.	
RC-10	Check active-loan counts after a renewal.	Deduplicated to the latest loan per member and loan type; source is `loan_schedules`, not `loan_status`.	

PART A2 · OTHER FLOWS

Membership, savings, CBU, ISC, termination, grocery

REGANT has several smaller lifecycles besides loans. Each moves a balance or a status and must reconcile the same way.

Membership

ID	TEST STEPS	EXPECTED RESULT	RESULT
MB-01	Applicant: Sign Up → Membership Form → attend training (Secretary records attendance) → Cashier records the membership fee → BOD approves.	Statuses follow Pending → Training → Approved → Official Member; a membership ID (TTMPC-…) is generated; a `member_account` row exists.	
MB-02	Confirm the same membership twice.	409 with the existing membership ID.	
MB-03	Approve an applicant with no training attendance record.	400 "no training attendance record found".	
MB-04	Record the membership fee / paid-up capital.	Receipt issued; it is linked to the member once they become a member.	
MB-05	Open a member's Personal Data Sheet with no PDS row.	Structured 404 with the next action.	

Savings

ID	TEST STEPS	EXPECTED RESULT	RESULT
SV-01	Cashier posts a savings deposit.	Validated at once; balance up; receipt issued.	
SV-02	Cashier records a withdrawal; Bookkeeper confirms it.	`pending_verification` → `validated`; ledger debit written; receipt issued only at confirmation.	
SV-03	Withdraw more than the balance.	400 insufficient funds.	
SV-04	Confirm the same withdrawal twice.	409.	
SV-05	Reject a pending withdrawal.	`rejected`; no ledger row; no receipt.	

Capital Build-Up (CBU)

ID	TEST STEPS	EXPECTED RESULT	RESULT
CB-01	Cashier records a CBU deposit at /Cashier_CBU_Deposit/:memberId.	New CBUD_nnn row (next number after the highest, not a count); share capital updated; receipt issued.	
CB-02	Deposit ₱0 / negative; deposit for a missing or terminated member.	400 / 404.	
CB-03	Double-submit a CBU deposit.	Credited once (WHAT_IFS: no dedup key yet: record behaviour).	
CB-04	Compare the member's share capital with the latest real CBU row.	Equal; the future-dated Dec-31 import placeholder is ignored.	

Interest on Share Capital (ISC)

ID	TEST STEPS	EXPECTED RESULT	RESULT
IS-01	Bookkeeper calculates ISC for a basis period starting before December 2025.	Refused.	
IS-02	Post a period overlapping an existing posting.	Refused (exclusion constraint).	
IS-03	Post; then at the March GA step (IscPayoutPreferencesModal) uncheck one member and settle.	Posting only records payables; settlement capitalises the checked members' ISC into CBU and marks the unchecked one as cash; a receipt per settled line.	
IS-04	Reverse a posting as the Bookkeeper who posted it, then as a Manager; reverse twice.	Only a Manager can reverse, never their own posting; a second reversal is refused.	

Account termination and CBU payout

ID	TEST STEPS	EXPECTED RESULT	RESULT
TM-01	Terminate a member whose CBU exceeds their loans.	CBU pays the loans first (system payments through `approve_bookkeeper_payment`); the rest appears for Cashier payout at /Cashier_CBU_Exit_Payout.	
TM-02	Terminate a member whose CBU is less than their loans.	The member stays active or becomes `exiting` (no new loans); auto-terminates when the balance reaches ₱0.	
TM-03	Cashier pays out the remaining CBU.	CBU ₱0 → member `closed`; receipt issued; personal contact fields cleared; login deleted; email replaced with closed-<id>@ttmpc.local.	
TM-04	Re-register with the closed member's old email.	Allowed (the auth user was deleted).	

Grocery / POS

ID	TEST STEPS	EXPECTED RESULT	RESULT
GR-01	Send a Completed POS event; resend the same event_id.	One GROCERY_TRANSACTIONS row; one receipt; the redelivery is a no-op, not an error.	
GR-02	Send a status other than Completed / On Credit; a negative amount.	400 to the POS side; no row (record the negative-amount behaviour: no CHECK yet).	
GR-03	Send an event whose member reference does not resolve.	The row is kept unmatched and visible to the Bookkeeper for linking, not silently dropped from totals.	

PART B · MONITORING

Dashboards, figures, and lists

Each portal has a dashboard (Member, Cashier, Bookkeeper, Manager, Treasurer, BOD) plus queue pages. Every figure must trace back to the rows behind it and stay inside the user's scope. Record which screen each result came from.

Figures and filters

ID	TEST STEPS	EXPECTED RESULT	RESULT
MO-01	For each headline figure (active loans, total CBU, savings, collections, pending approvals), count or sum the rows with a SQL query.	Equal; ₱ or count unit shown.	
MO-02	Change each filter (year, month, loan type, status).	Every figure and chart that claims to follow it changes; live queues and current balances stay and say so.	
MO-03	Choose a period with no loans.	"—" or "No records", not ₱0.00, unless 0 is real.	
MO-04	Check loans/payments dated on the first/last day of a month and Dec 31 / Jan 1.	Counted in the right period in Asia/Manila time.	
MO-05	Compare the Member dashboard figures with the Bookkeeper's view of that member.	Identical.	
MO-06	Check percentages (MIGS share, payment status share, delinquency rate).	Numerator/denominator traceable; no divide-by-zero; shares sum to 100% (±0.1).	
MO-07	Check year-on-year comparisons.	The compared period is named; arrow colours match meaning.	
MO-08	Check delinquency at 89, 90 and 91 days without payment; MIGS at 49 and 50.	Delinquent only from 90 days; MIGS from 50.	
MO-09	Switch tabs (e.g. SOA Loan · Savings · Transactions · CBU) and come back.	Filters kept or reset consistently.	
MO-10	Reset filters.	Back to the default (current year).	
MO-11	Check each queue count (pending loans, pending payments, pending withdrawals, override requests) against the list it opens.	Equal.	

Lists, search, and sorting

ID	TEST STEPS	EXPECTED RESULT	RESULT
MO-12	Page through long lists (Manage Members, Manage Loans, Audit Log, Receipts).	No row repeats or goes missing.	
MO-13	Sort by each sortable column.	Correct order for names, ₱ amounts (10 after 9), dates.	
MO-14	Search members by partial name, different case, "ñ", extra spaces, and by membership ID.	Matches found; clear "No results".	
MO-15	Search with % _ ' " < &.	No error; treated as text.	
MO-16	Combine search with filters, then remove one.	Results and count update.	
MO-17	Check "Showing x–y of n" counts.	Accurate.	

ID	TEST STEPS	EXPECTED RESULT	RESULT
MO-18	View a very long name or address in a table.	Wrapped or truncated with a way to read it.	

Drill-down and live updates

ID	TEST STEPS	EXPECTED RESULT	RESULT
MO-19	Click a dashboard card / chart / recent-activity row that links somewhere.	Opens the matching records with the same filters.	
MO-20	Open a loan from Manage Loans, then go Back.	Same page, scroll and filters.	
MO-21	As Cashier, record a payment while the Bookkeeper's /payments and dashboard are open.	Updates live via the `db-changes` realtime broadcast, without refresh.	
MO-22	Check the credit-risk queue and demand forecast for a model/last-updated note.	Model version or last-run time shown; stale data labelled.	
MO-23	Load the dashboards with a full year (743 legacy loans + new).	Loads within the agreed time; totals not cut off at 1,000 rows (Supabase default page size).	
MO-24	Maps.	N/A: REGANT has no maps.	
MO-25	Check active-loan counts after a member renews.	Counted once per member and loan type.	

PART C · VISUALIZATIONS

Charts and forecasts

REGANT's dashboards use Recharts. Run the universal checks once per screen with charts, check each chart's form, and fill the result column of the catalog.

Universal chart checks

ID	TEST STEPS	EXPECTED RESULT	RESULT
VZ-01	Hover, Tab-focus and tap each bar, point and slice.	Tooltip shows exact value, ₱/count unit and label.	
VZ-02	Read every value without hovering.	Values available as labels or a table.	
VZ-03	Compare chart totals with the table or report behind them.	Equal.	
VZ-04	Choose an empty period.	Empty-state message; no empty axes.	
VZ-05	Find a percentage based on very few records (e.g. a new year).	Labelled as too few records.	
VZ-06	Compare colours across portals.	The same status keeps the same colour (e.g. Paid, Pending, Overdue).	
VZ-07	Charts with many categories (loan types, transaction types).	Top items plus Other; no repeated colours.	
VZ-08	Charts mixing ₱ and counts.	Separate charts, never two y-axes.	
VZ-09	Axes.	Bars start at zero; ₱ and periods labelled.	
VZ-10	Time series (monthly collections, membership growth, approval trends).	Months continuous; missing months show as gaps.	
VZ-11	Pie/donut charts (gender, loan distribution, MIGS, payment status).	Add up to 100%; few parts.	
VZ-12	View in grayscale.	Meaning also given by labels.	
VZ-13	Click a chart item that filters, then click again.	Sets then clears the filter.	
VZ-14	Leave a chart open while data changes (realtime).	Updates without flicker.	
VZ-15	Open at phone width.	Charts fit; labels readable.	
VZ-16	Download the Bookkeeper Executive PDF.	Charts appear with the same values.	
VZ-17	Loan demand forecast card: compare forecast with actuals.	12 forecast months with the 80% interval shaded; actuals and forecast clearly distinguished; Consolidated and Emergency separate.	
VZ-18	Credit-risk display: thresholds and metric shown.	Bands come from `/api/risk/model-info`; any quoted performance is AUC 0.682 (not 0.907).	

Chart catalog

CHART	WHERE	WHAT IT SHOWS	FILTERS IT FOLLOWS	RESULT
Bar	BOD Dashboard	Approved loans per month	Year	

CHART	WHERE	WHAT IT SHOWS	FILTERS IT FOLLOWS	RESULT
Bar / list	BOD Dashboard	Cooperative transactions	Period	
Pie	BOD Dashboard	Gender distribution of members	—	
Bar	Bookkeeper Dashboard	Repayment behaviour	Year	
Bar	Bookkeeper Dashboard	Yearly collections	Year	
Pie	Bookkeeper › Reports	Loan distribution by type	Report period	
Line	Bookkeeper › Reports	Membership growth	Report period	
Bar	Bookkeeper › Reports	Monthly collections	Report period	
Pie	Bookkeeper › Reports	MIGS classification	Report period	
Pie	Bookkeeper › Reports	Payment status	Report period	
Area	Cashier Dashboard	Transaction volume	Period	
Pie	Cashier Dashboard	Transaction breakdown by type	Period	
Area	Manager Dashboard	Loan approval trends	Period	
Pie	Manager Dashboard	Loan distribution	Period	
Line + band	LoanDemandForecastCard	12-month SARIMAX forecast, 80% CI	Loan type	
Score / bands	Credit Risk (Bookkeeper, Manager)	Default probability per application	Queue filters	
(fill in)	Manager Reports			

Is it the right chart?
Use the generic guide's form table. Examples for REGANT: a single total CBU figure is a number tile, not a chart; loan type mix with ≤ 6 types is a donut; collections per month is columns; the approval funnel (pending → recommended → approved → released) would be a funnel.

PART D · INPUTS AND OUTPUTS

Every user input and data output

Every input must accept what it should and refuse everything else, in the browser and on the server (FastAPI Pydantic models, Postgres CHECKs, RLS). Every output (loan rows, receipts, audit log, bell notifications, Resend emails, PDFs) must say the same as the screens.

Input validation by REGANT field

FIELD	TRY THESE VALUES	EXPECTED
Names, address	Empty; spaces only; ñ, é; 200+ chars; <script>alert(1)</script>	Required rules; trimmed; shown as text in screens and PDFs.
Email	Missing @; capitals; name+qa@...; an email already used	Invalid refused; uniqueness ignores case.
Mobile	09171234567, +639171234567, letters, 10 digits	PH formats accepted; stored consistently.
TIN	9 and 12 digits, dashes, letters	Formatted consistently (used as co-maker ID).
₱ amounts (loan, payment, deposit, withdrawal, CBU)	0; −1; 0.001; 1,000.50; 1.000,50; above caps	> 0 enforced on the server; 2 decimals; caps (Emergency ₱20,000; ceiling 5×/3× CBU) enforced.
Term	6, 12, 18, 24, 60, 72	Only the type's allowed terms.
Dates (birth, payment, basis period)	Today; future; 29 Feb 2027; 31 Apr; before Dec 2025 (ISC)	Invalid refused; future payment dates handled as documented.
Reference number (release)	Empty; duplicate	Required; duplicate → 409.
Remarks (reject / revise / reschedule)	Empty; 5,000 chars	Required where documented.
Resolution no./date (BOD)	Empty on approve	422.
Dropdowns (loan type, collateral type, role)	An inactive or out-of-list value sent by editing the request	Refused by CHECK / server.
File upload (membership form attachments, if used)	.exe renamed .jpg; over limit; 0 bytes	Type checked; request body > 1 MB → 413.
Password	7 chars; mismatch	Min 8; never shown, emailed or logged.

Input test cases

ID	TEST STEPS	EXPECTED RESULT	RESULT
IN-01	Submit the Consolidated form with several errors.	All errors shown next to their fields; entries kept.	
IN-02	Read each error message.	Says what is wrong and how to fix it; no stack traces or raw 422 JSON.	
IN-03	Bypass the browser checks (Supabase client or API client).	The server/DB enforces the same rules.	
IN-04	Use the loan form with the keyboard only.	Logical order; Enter submits once; Esc closes modals.	
IN-05	Use autofill and paste.	Accepted; validation still runs.	
IN-06	Check required markers.	Every required field is marked.	

ID	TEST STEPS	EXPECTED RESULT	RESULT
IN-07	Change loan type or term.	Amount ceiling, rate, fees and amortization update; stale values cleared.	
IN-08	Leave a long form (Membership Form, PDS) halfway.	Draft restored or loss warned.	
IN-09	Fill ₱ fields on a phone.	Numeric keyboard.	
IN-10	Open a new form.	Sensible defaults (today's date; the signed-in member).	

Data outputs and where to verify them

OUTPUT	WHERE TO VERIFY IT	WHAT MUST BE TRUE
Loan rows and statuses	Manage Loans, Loan Approval, Member Loans, DB	Match the last action.
Balances and ledgers	SOA, Loan Ledger, Savings Details, CBU, Vault	Opening + in − out = closing.
Audit log	Each role's audit-log page; /audit-trail	Actor, action, record, time, change.
Bell notifications	`loan_notifications` bell feed	Right recipients; cleared when read.
Emails	Test inbox (Resend)	Right recipient; links point to the test URL; no secrets.
Receipts	Pop-up, SOA Transactions tab, staff Receipts pages	TTMPC-TR-YYYY-NNNNNN, unique and sequential; frozen snapshot; footer "This is not a BIR Official Receipt".
Generated PDFs	Loan forms (src/PDF templates), SOA, savings/CBU statements, Executive PDF	Values match the record; Consolidated > ₱500,000 uses the BOD template.
Integrations	POS webhook; FastAPI logs	Complete; failures visible.

Output test cases

ID	TEST STEPS	EXPECTED RESULT	RESULT
OU-01	Create, approve, release, pay a loan; open the audit log.	One entry per action with actor, time, record, change.	
OU-02	As any staff, try to edit/delete audit-log entries (screen and Supabase).	No control; refused.	
OU-03	Check who gets each bell notification and email.	Only the member and the next actor; not the actor themselves.	
OU-04	Open links in emails from the test environment.	Point to the test URL; expired links show a clear message.	
OU-05	Check receipt numbers across a year boundary and after a failed transaction.	Unique, never reused; restart per year as documented.	
OU-06	Compare ₱ and date formats on screen, PDFs, receipts and emails.	Same everywhere (₱1,234.50).	
OU-07	Remove RESEND_API_KEY (with the dev team) and approve a loan.	Approval still saves; email returns 503 "Email disabled" and is logged.	
OU-08	Try to update or delete a receipt row in the DB.	Refused: "Receipts cannot be edited or deleted".	

ID	TEST STEPS	EXPECTED RESULT	RESULT
OU-09	Run the receipt verification queries (`src/server/receipt_checks/`).	Every "should be 0" check is 0; hash mismatches 0.	

Reports and exports

ID	TEST STEPS	EXPECTED RESULT	RESULT
RP-01	Download the Executive PDF, SOA (Loan Ledger), member SOA / savings / CBU PDFs, receipts.	Each opens without errors.	
RP-02	Compare report figures with the screen.	Identical.	
RP-03	Download a member's SOA as that member.	Only their data.	
RP-04	Download for a period with no data.	Says no data, or the option is disabled.	
RP-05	Download the largest period (all years).	Finishes; no missing rows.	
RP-06	Check headers.	Title, period, generated time, member/prepared-for.	
RP-07	Check numbers and dates.	Correct ₱ and dates.	
RP-08	Names with ñ / é.	Shown correctly in PDFs.	
RP-09	CSV/Excel formula injection.	N/A unless an Excel/CSV export exists: REGANT exports PDFs only (verify).	
RP-10	Check file names.	Name the report, member and period.	
RP-11	Print a PDF in grayscale.	No cut-off columns; page numbers; readable.	
RP-12	Download on a throttled connection.	Clear message; no partial file.	

PART E · CROSS-CUTTING

Shared behavior across the system

Behaviour every REGANT screen shares.

Dialogs, feedback, and shared tools

ID	TEST STEPS	EXPECTED RESULT	RESULT
XC-01	Start each destructive action (reject loan, reject payment, terminate, reverse ISC).	Confirmation names the record, uses a danger style; Cancel changes nothing.	
XC-02	Save/submit on each form.	Loading state; success message; clear failure message; button disabled.	
XC-03	Watch for native alert/confirm/prompt boxes.	None; REGANT modals used.	
XC-04	Trigger bell notifications.	Appear, clear when viewed; not for the actor's own action.	
XC-05	In-app messages.	N/A unless a chatbot/messaging feature ships (see CHATBOT_PLAN).	
XC-06	Calendars (General Assembly, payroll release schedule).	Correct dates in Asia/Manila; delayed cycles shown as pending, not omitted.	
XC-07	Help text and empty states.	Explain the next step.	
XC-08	Unknown URL; API error; page without permission.	Unknown URLs redirect to the landing page; no stack traces.	
XC-09	Terms across screens (CBU / Share Capital, Released / Disbursed).	Consistent wording.	
XC-10	Name, logo and colours on screens, emails, PDFs, receipts.	Consistent TTMPC/REGANT branding.	

Responsive layout

ID	TEST STEPS	EXPECTED RESULT	RESULT
RL-01	View each main screen at 360, 768, 1024, 1440 px.	No sideways page scroll.	
RL-02	Member portal bottom navigation on a phone.	Opens the right sections; current one shown.	
RL-03	Wide tables on a phone (SOA, Loan Ledger).	Scroll inside their box.	
RL-04	Tap targets on a phone.	About 44×44 px.	
RL-05	Rotate while filling the loan form.	No input lost.	
RL-06	Zoom to 200%.	Readable and usable.	

Accessibility

ID	TEST STEPS	EXPECTED RESULT	RESULT
AX-01	Keyboard-only through the loan application and approval.	Every control reachable; no focus trap.	
AX-02	Focus indicator.	Always visible.	
AX-03	Screen reader (NVDA) on login and loan form.	Meaningful names; errors announced.	

ID	TEST STEPS	EXPECTED RESULT	RESULT
AX-04	Form labels.	Visible labels; not placeholder-only.	
AX-05	Contrast.	≥ 4.5:1 text, 3:1 large text/icons.	
AX-06	Images and icons.	Alt text where meaningful.	
AX-07	Reduced motion.	Animations reduced.	
AX-08	Page titles and headings.	Unique titles; logical headings.	

Browsers and devices

ID	TEST STEPS	EXPECTED RESULT	RESULT
BC-01	Smoke flow (sign in, apply, recommend, approve, release, report) in Chrome and Edge.	Works; versions recorded.	
BC-02	Outdated browser.	Clear message or basic function.	
BC-03	File picker / camera on Android and iOS (membership form).	Works or explains denial.	
BC-04	Install the PWA (vite-plugin-pwa), then deploy an update.	Installs, opens, moves to the new version.	

Performance

ID	TEST STEPS	EXPECTED RESULT	RESULT
PE-01	Load the landing page and each dashboard.	< 3 s on broadband, < 6 s on mobile data.	
PE-02	Save and submit.	< 2 s or a progress indicator.	
PE-03	Largest lists, Reports, demand forecast (cold start).	No timeouts.	
PE-04	Several staff working at once (Cashier + Bookkeeper + Manager).	No errors or duplicate rows.	
PE-05	Leave a realtime dashboard open for an hour.	Does not slow down.	
PE-06	Throttle to Slow 4G.	Loading states; nothing submitted twice.	
PE-07	Send > 100 API requests per minute from one IP.	429 from the rate limiter.	

Network and offline

ID	TEST STEPS	EXPECTED RESULT	RESULT
NW-01	Drop the network while recording a payment / releasing a loan; retry.	Error shown; no duplicate or lost record.	
NW-02	Work offline.	N/A: REGANT has no offline mode; actions say they need a connection.	
NW-03	Reconnect after offline.	N/A: no offline queue.	
NW-04	Session expires while offline.	Sign in again; unsaved form input warned.	

Security basics

Not a penetration test
Basic checks only. See also the known risks in AC-19 and AC-21 and the System Evaluation notes.

ID	TEST STEPS	EXPECTED RESULT	RESULT
SE-01	Open the deployed site over http://.	Redirects to HTTPS.	
SE-02	Developer tools on each screen: source, network, storage.	No service-role key, Resend key or other members' data; only the Supabase anon key.	
SE-03	Request another member's loan/savings/receipt by ID.	Refused (RLS / 404).	
SE-04	<script>alert(1)</script> in names, remarks; view everywhere incl. PDFs.	Shown as text.	
SE-05	Upload a disguised file.	Refused; never served as a page.	
SE-06	Many failed sign-ins / rapid requests.	Slowed or limited (Supabase Auth; 429).	
SE-07	Check session storage.	Tokens in sessionStorage (by design); note the XSS exposure in A4.	
SE-08	Cause errors on purpose.	No stack traces, SQL or file paths.	
SE-09	Password-reset codes.	Single use, expiring; no account enumeration.	
SE-10	Sign out, then replay an earlier API request with the old token.	Refused (or record that the JWT stays valid until expiry).	
SE-11	Screens and PDFs with personal data (PDS, member details).	Only roles that need them; Secretary records view-only.	
SE-12	Review defect evidence.	No passwords, tokens, reset codes.	

Data integrity

ID	TEST STEPS	EXPECTED RESULT	RESULT
DI-01	Two Cashiers post to the same member's savings at once.	Final balance correct.	
DI-02	Cause a failure mid-release (with the dev team).	All or nothing: no released loan without schedule or receipt.	
DI-03	Delete a member with loans (DB).	Blocked or kept consistent; collateral cascade matches intent.	
DI-04	Legacy data import / backfill (`/api/admin/backfill-member-auth`).	Valid rows processed; invalid ones reported.	
DI-05	Restore a Supabase backup to a branch.	Complete and consistent.	
DI-06	Retention: closed members.	Financial history and DOB kept; contact data cleared.	
DI-07	Every table touched by tests has RLS policies (Supabase linter `rls_enabled_no_policy`).	No table silently returns empty results.	

Time, locale, and language

ID	TEST STEPS	EXPECTED RESULT	RESULT
TL-01	Act at 23:59 and 00:00 Asia/Manila (payment, CBU deposit).	Dated to the correct day.	
TL-02	Month, year ends; 29 Feb; Bonus window at Apr 30 → May 1 and Oct 31 → Nov 1.	Totals correct; the window opens/closes at Manila midnight.	
TL-03	Change the computer's time zone.	Times shown consistently.	
TL-04	₱ and date formats.	Philippine format, consistent.	
TL-05	Switch language.	N/A: English only.	

APPENDICES

Templates and reference

Filled in for REGANT from the code. Confirm any row marked “verify” on the test build.

A1. Status map

RECORD	STATUS (EXACT)	MEANING	ALLOWED NEXT	WHO
Loan	pending	Submitted; awaiting Bookkeeper	recommended for approval, recommended for bod approval	Bookkeeper
Loan	recommended for bod approval	Consolidated/KOICA > ₱500,000	recommended for approval, bod rejected	BOD
Loan	recommended for approval	Awaiting Manager	to be disbursed, rejected, revision_requested	Manager
Loan	revision_requested	Returned to Bookkeeper	recommended for approval (resubmit)	Bookkeeper
Loan	to be disbursed	Approved; awaiting Treasurer	ready for disbursement, pending rescheduling, rejected	Treasurer
Loan	pending rescheduling	Parked by Treasurer	rescheduled → to be disbursed (verify)	Treasurer / Manager
Loan	ready for disbursement	Cleared; awaiting Cashier	released	Cashier
Loan	released	Money out; schedule active	fully paid; (renewal pays it off)	System (payments)
Loan	fully paid	Balance ₱0	— (terminal)	—
Loan	rejected / bod rejected	Ended	— (terminal)	—
Loan payment	pending_bookkeeper	Recorded by Cashier	validated, rejected	Bookkeeper
Schedule row	Unpaid / Partially Paid / Fully Paid	Instalment state	forward only	System
Savings txn	pending_verification	Withdrawal recorded	validated, rejected	Bookkeeper
Membership application	Pending → Training → Approved → Official Member / Rejected	Applicant journey	forward only	Secretary, Cashier, BOD
Member	active / exiting / terminated / closed	Account lifecycle	active→exiting/terminated→closed	Account-admin BOD; system
Renewal override	pending / approved / rejected / consumed / expired (verify names)	6-month rule waiver	pending→approved/rejected; approved→consumed/expired	Bookkeeper (not own)
Restructure request	pending / approved / rejected / cancelled	Term change	pending→one of the others	Manager
ISC transaction	payable → capitalised / cash; reversed	Dividend line	settle once; reverse once	Bookkeeper post; Manager reverse
Grocery txn	Completed / On Credit	POS sale	—	POS

A2. Business rules and limits

RULE	VALUE	WHERE ENFORCED	TEST CASE
Consolidated interest	0.83%/month add-on	Compute endpoint / form	RC-01
Consolidated terms	12, 24, 36, 48, 60 months	Pydantic validator	PF1-07
Emergency cap / terms / rate	₱20,000; 6 or 12 months; 2% diminishing	Pydantic validator	PF1-07
Bonus window	May and November only; DepEd only	DB trigger	PF1-08, TL-02
BOD approval threshold	Consolidated/KOICA > ₱500,000 (exclusive)	LoanApprovalDetails.jsx; PDF template	PF2-02
Loan ceiling	CBU × 5 (MIGS) / × 3 (Non-MIGS)	Loan forms / eligibility	PF1-01
MIGS threshold	≥ 50 of 100 (7 criteria)	migs_engine.py	PF0-05, MO-08
Renewal rule	6 validated payments (RENEWAL_MIN_PAYMENTS)	get_loan_eligibility + override	PF8-01/02
Override validity	30 days or one renewal	Renewal override endpoints	PF8-03
Penalty	2%/month (Bonus 1%) on remaining principal after 3-month grace; none on legacy loans	Penalty accrual	PF7-07
Delinquency	90 days with no payment	Reports / dashboards	MO-08
ISC basis period	Cannot start before December 2025; no overlaps	isc_calculate_preview / constraint	IS-01/02
Request limits	100 requests/min per IP; 1 MB body	slowapi; body-size middleware	PE-07
Password	Minimum 8 characters	Reset / sign-up forms; Supabase	AC-12
Receipt number	TTMPC-TR-YYYY-NNNNNN	receipt_next_no()	OU-05

A3. Boundary values cheat sheet

INPUT	VALUES TO TRY
High-value loan	₱499,999.99 · ₱500,000.00 · ₱500,000.01
Emergency amount	₱19,999.99 · ₱20,000.00 · ₱20,000.01
Loan ceiling	Exactly CBU × 5 (or × 3), ₱0.01 more, ₱0.01 less
MIGS score	49 · 50 · 51
Renewal payments	5 · 6 · 7 validated payments
Delinquency	89 · 90 · 91 days
Penalty grace	Day before and after 3 months past due
Bonus window	Apr 30 23:59 · May 1 00:00 · May 31 23:59 · Jun 1 00:00 (and Oct/Nov)
Withdrawal	Balance − 0.01 · balance · balance + 0.01
Amounts	0 · 0.01 · −0.01
Override age	Day 29 · 30 · 31

A4. Documentation differences to confirm

#	DOCUMENT AND SECTION	WHAT IT SAYS	WHAT THE SYSTEM DOES	DECISION
1	DEMO_GUIDE.md §7	Credit risk is Logistic Regression via POST /score-loan with Stability_Score etc.	v3.3 HistGradientBoosting, 21 features, POST /api/risk/predict, AUC 0.682	
2	DEMO_GUIDE.md checklist / §8	GET /demand-forecast; /send-loan-status-email	/api/analytics/demand/forecast; /api/send-status-email	
3	WHAT_IFS · Loans	Loan status transitions not guarded server-side	Generic guide expects server refusal (AC-19, PF4-07)	
4	WHAT_IFS · Staff termination	Session revocation on deactivation unspecified	Termination bans the auth user (AC-15)	
5	Loan policies (printed)	Paper workflow	REGANT intentionally changes the workflow (thesis scope): not a defect by itself	
6	ISC_DIVIDEND_PLAN.md §12/§14	Rate derived from audited surplus; cash by default with capitalise election	Verify the build matches before posting	
7	—	No documented sign-in lockout or idle timeout	Record observed behaviour (AC-03, AC-13)	

A5. Defect report template

FIELD	WHAT TO WRITE
Defect ID / Test case ID	e.g. D-014 / PF6-02
Summary	One line: what is wrong and where.
Severity	Critical, High, Medium, Low.
Role and account	Role and test account (e.g. Cashier, QA Cashier 01); member ID if relevant.
Location	Portal › page › tab and the route (e.g. /Cashier_Disbursement).
Date and time	Asia/Manila.
Record identifier	Loan control number, membership ID, receipt number, CBUD id.
Steps to reproduce	Numbered, from a known state.
Expected / Actual	As in this guide / what happened, with the exact message.
Evidence	Screenshot, recording, downloaded PDF, API response (no tokens).
Device and browser	e.g. Windows 11, Chrome 140.

Never include
Passwords, reset codes, JWTs, the Supabase service-role key, Resend key, or members' personal data beyond what is needed.

APPENDICES

Test summary and sign-off

Case counts are computed from this guide.

A6. Test summary and sign-off

AREA	CASES	PASS	FAIL	BLOCKED	N/A
Access and roles	22				
Process flow stages 0–8	60				
End-to-end reconciliation	10				
Other flows	25				
Monitoring	25				
Visualizations	18				
Inputs	10				
Outputs	9				
Reports and exports	12				
Dialogs and shared tools	10				
Responsive layout	6				
Accessibility	8				
Browsers and devices	4				
Performance	7				
Network and offline	4				
Security basics	12				
Data integrity	7				
Time, locale, and language	5				
Total	**254**				

ROLE	NAME	SIGNATURE	DATE
QA assessor			
QA lead			
Development lead			
TTMPC representative / thesis adviser			

Build tested: _____ Environment: _____ Test period: _____

A7. Glossary

TERM	MEANING
CBU	Capital Build-Up: the member's share capital.
MIGS	Member In Good Standing: score ≥ 50 of 100; raises the loan ceiling to 5× CBU.
Bona fide member	A member eligible for Consolidated and Bonus loans.
ISC	Interest on Share Capital: the annual dividend, capitalised or paid in cash.
KOICA / ABFF	The non-member loan programme.
Release	The Cashier pays out an approved loan (status released).
Renewal	A new loan that pays off the old one (new row).
Restructure / reschedule	Same loan, new term or schedule.
Legacy loan	One of the 743 imported loans; no penalties and no receipts.
Receipt	System-generated transaction receipt, not a BIR Official Receipt.
Smoke / regression test	Quick check that every role opens its home screen / rerunning cases after a fix.
Idempotent	Same effect whether it runs once or twice (e.g. a payment cannot be validated twice).